Spectra Sleuth Showdown

Can two white lights hide different color fingerprints?

Win condition: Most correct spectrum matches and strongest safe fireworks-science explanation.

THE SCIENCE YOU ARE USING

Diffraction splits light. A diffraction grating bends different colors by different amounts, spreading light into a spectrum. Two lights that look the same white can have very different spectra: a smooth rainbow from a hot filament, or separated bright lines from an LED or gas.

Spectra as fingerprints. Each light source has a characteristic spectrum, its fingerprint. This is the same science behind firework colors, where different metal salts emit specific colors. You match mystery sources to clue cards using their spectra, no flame needed.

HOW TO BUILD AND RUN IT

1. **Look through the grating.** View a white light through the grating and watch it spread into colors.
2. **Compare two sources.** Look at an LED and a filament bulb. Notice how their spectra differ.
3. **Sketch the spectra.** Draw what you see for each mystery source: continuous band or separate lines.
4. **Match to clue cards.** Use your sketches to match each source to its spectrum card.
5. **Connect to fireworks.** Explain how this links to the colors in a firework display, safely.

Safety: No flame tests. No lasers. Diffraction glasses are not eye protection. Allergy: None.

MATERIALS

Diffraction glasses or gratings	per student
LED sources	shared
Incandescent source	if available
Spectrum cards	set
Museum clue sheets	per team

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why):

The ONE variable we are testing:

Baseline result:

After redesign:

DEFEND YOUR DESIGN

What evidence made you change your design?

If you ran it again, what is the ONE thing you would change?

HOW YOU SCORE (100 POINTS)

Scoring criterion	Points
Correct matches	**40**
Spectrum sketches	**20**
Explanation	**20**
Exhibit connection	**20**
Total	**100**